YUAN ZE UNIVERSITY

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ENERGY AS A CONSTRAINT

Credibility, Pricing, and Settlement in Energy-Linked Digital Finance

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of Master of Science in Finance

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Abstract

Can energy discipline digital financial claims, or does it merely provide a physical narrative for value determined elsewhere? This thesis treats that question as a problem of constraint design. It combines an empirical test of Bitcoin mining-cost valuation, a renewable-energy risk benchmark, and an executable architecture for claim admission, quantity, and settlement.

The empirical analysis reconstructs the Cumulative Energy Investment Ratio (CEIR) using a daily Bitcoin panel from 2019 to 2025. The apparent pre-split association is statistically significant ($\beta = -0.2623$; HAC $p = .00052$), but it does not survive identification. Removing the pre-sample cumulative-cost stock weakens the coefficient to -0.1037 ($p = .099$), and the robust joint break test is insignificant ($p = .133$). CEIR is also almost perfectly correlated with market capitalisation divided by cumulative energy use ($r = .999989$), while a cumulative-time control produces a similar result. Passive mining-cost ratios therefore do not identify a clean energy-specific value anchor.

The constructive analysis replaces that failed shortcut with explicit financial rules. An option-style scenario framework converts declared renewable-output uncertainty into inspectable pricing, margin, and oracle-tolerance quantities. A Sepolia proof and deterministic case workbench then separate evidence admission, compatible quantity ceilings, settlement capacity, and decision provenance. Controlled cases show that a claim may be blocked for weak evidence, limited by different constraints, or admitted yet later settle only partially.

The thesis contributes a five-constraint architecture: reliable energy evidence, rule-bound issuance, explicit pricing and risk controls, protected settlement and redemption accounting, and limited governance. Its central result is conditional: energy can discipline a digital financial claim only when these constraints operate jointly and remain auditable. The implementation establishes research feasibility under controlled evidence and model-based context, not production or legal readiness.

Keywords: Energy-linked finance, digital money, monetary credibility, renewable energy risk, Bitcoin energy cost, decision systems, smart contracts, reproducible research

JEL Codes: E42, G13, Q42, Q47

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Chapter 1 - Introduction

1.1 Research Motivation

Bitcoin and renewable energy expose opposite weaknesses in attempts to connect physical resources with digital finance. Bitcoin links digital scarcity to real electricity expenditure, yet holders receive no claim on the electricity consumed. Renewable generation produces usable electricity, yet production alone does not create a credible financial claim. The central problem is therefore not whether energy is real, but whether an energy relationship can impose a verifiable limit on what a financial system may promise.

Monetary systems have historically relied on different forms of discipline. Gold-linked systems used physical scarcity and convertibility. Fiat money relies on institutions, law, and policy credibility, while Bitcoin relies on protocol rules and costly proof-of-work (Barro & Gordon, 1983; de Vries, 2018; Eichengreen, 1992; Friedman, 1960; Nakamoto, 2008). Each form constrains one margin while leaving another exposed. Gold creates custody and redemption pressure, fiat preserves flexibility at the cost of discretion, and Bitcoin creates costly scarcity without redemption rights.

Energy appears to offer what digital systems often lack: a physically observable production reference. It is also difficult to translate into a uniform financial constraint. A kilowatt-hour varies in value with time, location, grid access, storage, curtailment, and settlement terms (International Energy Agency, 2023; Joskow, 2011; Lazard, 2025; Ueckerdt et al., 2013). Production potential, measured output, contractual value, and settled delivery are related but non-equivalent facts.

This thesis asks whether verified energy evidence can constrain claim creation more directly than passive expenditure does. The proposed answer is institutional as well as technical: energy matters only when evidence, issuance, pricing, settlement, and governance jointly restrict the claim.

1.2 Problem Statement

Energy-linked finance can appear constrained while remaining economically discretionary. A fixed token supply does not reveal whether the underlying evidence is reliable, whether the permitted quantity is defensible, or whether settlement obligations can be met. An energy reference can therefore coexist with weak data, arbitrary issuance, underpriced risk, and rules that can be revoked when inconvenient.

Bitcoin provides the empirical test of the first shortcut: that cumulative energy expenditure itself anchors value. Reproducing a mining-cost ratio is not enough. An energy-specific interpretation requires the relationship to survive alternative cumulative denominators, changes to the initial cost stock, time-series diagnostics, and robust break tests.

Renewable energy presents a second shortcut. It creates productive output, but the output and its financial value vary across locations and states of the world. A model can expose that uncertainty, yet a calculated price or risk measure constrains nothing until policy determines how it changes admission, quantity, margin, or settlement protection.

The research problem is therefore to identify and implement the conditions under which energy evidence actually limits a claim. A credible system must state what evidence counts, when a claim may exist, how much may be created, how uncertainty is absorbed, what is owed at settlement, and who may alter those rules.

Without those answers, 'energy-linked' describes a narrative rather than a financial constraint.

1.3 Research Question

The thesis addresses one primary question:

> *Under what conditions can verified energy evidence impose a credible and enforceable constraint on digital financial claims?*

The question is deliberately narrower than whether energy should become money. It asks whether an energy relationship can reduce discretion over claim creation, quantity, and settlement.

Three supporting questions test that proposition:

1. Can a Bitcoin mining-cost ratio identify a valuation effect that is specific to energy cost rather than persistence, cumulative construction, or time?
2. How can renewable-energy uncertainty be translated into explicit pricing, margin, and data-tolerance quantities?
3. Can evidence quality, admission, quantity limits, settlement capacity, and policy lineage be represented as separate, reproducible decision stages?

Together, the questions follow a cumulative logic: falsify a passive energy anchor, translate uncertainty into explicit financial quantities, and test whether those quantities and evidence rules can govern an executable claim process.

1.4 Research Design

The research design proceeds through three connected layers: empirical diagnosis, financial translation, and executable constraint.

The empirical layer asks whether market value relative to cumulative Bitcoin mining electricity cost predicts later returns and, more importantly, whether any relationship is truly attributable to energy prices. CEIR is the test ratio. Negative controls replace cost with cumulative energy use and cumulative time; sensitivity checks remove the starting cost stock; and the 2021 China mining-ban period is treated as a candidate break. These tests address structural-break, persistence, overlapping-return, and ratio-construction concerns (Chow, 1960; Granger & Newbold, 1974; Hodrick, 1992; Kronmal, 1993).

The financial layer asks how uncertain renewable-energy output can be translated into inspectable risk quantities. A transparent cold-start benchmark links declared variability to option values, collateral, and data tolerance without claiming to discover a traded market price (Bessembinder & Lemmon, 2002; Lucia & Schwartz, 2002).

The implementation layer tests whether the proposed constraints can govern actual decision stages. The Sepolia proof covers selected issuance and settlement rules. The case workbench handles the prior decision: it records evidence and context, applies a declared policy, determines whether a claim may proceed, limits its quantity, and preserves a reproducible receipt. The two layers establish feasibility under controlled assumptions, not production readiness.

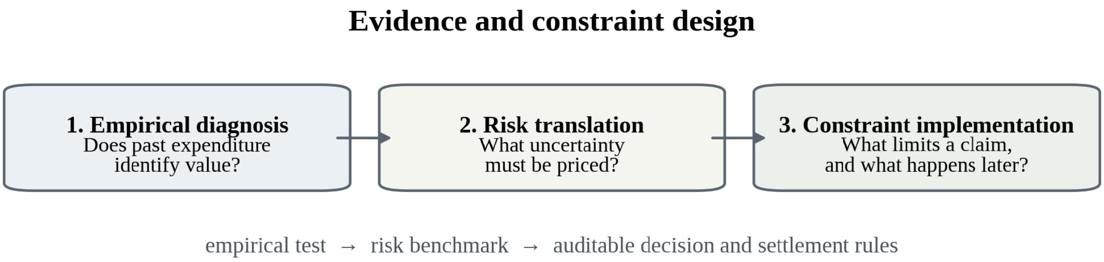

Figure 1.1. Three-part evidence design.

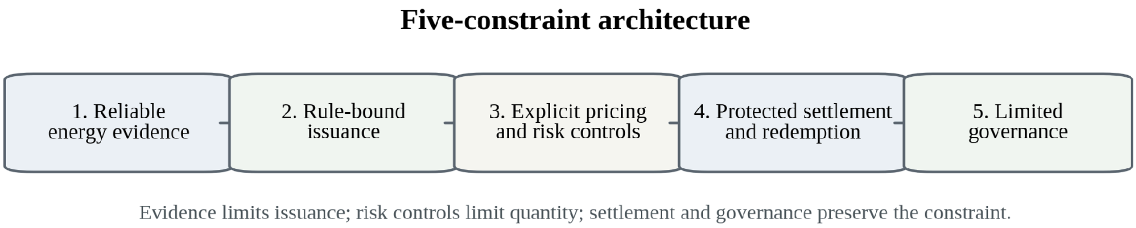

Figure 1.2. Integrated constraints for an energy-linked financial claim.

1.5 Contributions

The principal contribution is an integrated architecture for credible energy-linked digital finance. Real energy expenditure or production is insufficient on its own; reliable data, rule-bound issuance, explicit treatment of risk, protected settlement, and limits on rule changes must operate together.

Conceptually, the thesis positions energy between three familiar models of monetary discipline. Gold provides a physical asset, fiat relies on institutional credibility, and Bitcoin creates protocol scarcity. Energy may provide a verifiable production constraint without copying gold convertibility or treating proof-of-work expenditure as intrinsic value.

Empirically, the thesis contributes a negative identification result. The original CEIR regression reproduces a significant pre-split association, but the relationship is not unique to electricity cost, is sensitive to the starting cumulative-cost stock, and is unsupported by the preferred robust break test. Bitcoin therefore identifies the limit of passive anchoring rather than validating an energy-backed currency.

Methodologically, the pricing chapter provides a transparent route from declared renewable-output uncertainty to option-style values, margin stress, and oracle-tolerance measures. The contribution is not a universal price; it is an inspectable way to expose the assumptions that a claim would otherwise hide.

Architecturally and technically, the implementation separates evidence admission, quantity limits, and later settlement. Sepolia shows that selected issuance and settlement-accounting rules can execute on a public testnet. The case workbench shows why a claim is blocked, what binds its quantity, and whether it later settles in full. Decision identity and evidence-policy lineage remain reproducible.

The broader contribution is integration. The thesis connects monetary credibility, Bitcoin energy-cost identification, renewable-energy risk, pricing, institutional design, smart contracts, and reproducible decision systems within one argument about constraint.

1.6 Scope of the Thesis

The claims are deliberately bounded. The thesis does not propose energy as a replacement for fiat money, assume that an energy-linked token is safe, or present the proposed system as legally or commercially ready. Bitcoin is not treated as energy-backed, and public resource models are not treated as proof of site-level production.

Three boundaries define the evidence:

1. The Bitcoin analysis identifies the limits of one mining-cost ratio. It is neither a general law of digital-asset valuation nor evidence of an energy-price floor.
2. The renewable-energy model is a cold-start framework for exposing pricing and risk assumptions. It is not a complete electricity-market model, a traded price, or a legal settlement mechanism.
3. The implementation is proof-of-concept research. The Sepolia contracts use a public testnet, while the case workbench uses controlled evidence and model-based PVWatts/TMY context. Neither provides operator validation, authority to create production claims, or evidence of deployment readiness.

Legal classification, consumer protection, liquidity, reserve custody, production cybersecurity, utility regulation, tax, and long-term governance remain outside the main scope. The thesis addresses the prior design question: what must be constrained before an energy-linked financial claim can be considered credible at all?

Chapter 2 - Literature Review and Theoretical Background

2.1 Credible Constraint as the Organizing Question

The literature does not lack explanations for scarcity, credibility, production risk, or automated enforcement. It lacks a common account of how those elements must interact before a physical reference can limit a digital financial claim.

Monetary economics explains why commitment and limits on discretion matter (Barro & Gordon, 1983; Kydland & Prescott, 1977). Gold and Bretton Woods show how a physical reserve becomes credible only through convertibility and settlement pressure (Bordo, 1993; Eichengreen, 1992; Federal Reserve History, 2013). Bitcoin shows that protocol scarcity can make digital creation costly without giving holders a claim on the consumed resource (de Vries, 2018; Nakamoto, 2008).

Renewable-energy finance adds productive but variable output (International Energy Agency, 2023; Lazard, 2025). Pricing theory forces uncertainty into explicit assumptions, while smart contracts can enforce declared rules (Black & Scholes, 1973; Cong & He, 2019; Cox, Ross, & Rubinstein, 1979). Each literature solves only part of the design problem.

The organising question is therefore not which literature supplies the correct metaphor for 'energy money'. It is which combination of commitment, evidence, pricing, settlement, and governance turns an energy reference into a credible constraint.

2.2 Monetary Credibility, Rules, and Discretion

The first requirement is commitment. Kydland and Prescott (1977) show that a rule announced today may be abandoned after users have acted on it. Credibility therefore depends not only on the current rule but also on the cost and visibility of changing it later.

Barro and Gordon (1983) identify the related expectations problem: promises are discounted when users expect future discretion. In digital finance, a claim of limited supply, reserves, or energy linkage remains weak if administrators can mint outside the rule, replace the evidence source, revise the pricing method, or avoid redemption.

This literature yields a design principle rather than a specific asset choice: the relevant limits must bind the issuer as well as the user. Data acceptance, issuance, pricing, settlement, and governance must be observable and difficult to override selectively.

The remaining question is what gives those limits economic substance. Gold, fiat money, and Bitcoin answer that question differently, and their failures reveal why no single form of scarcity is sufficient.

Table 2.1 compares these sources of discipline and the failure mode attached to each.

Table 2.1. Sources of discipline across monetary systems.

System	Main Constraint	Main Strength	Main Failure Mode
Gold-backed money	Physical scarcity and redemption into gold	Hard to create gold from nothing	Custody, verification, physical settlement, and redemption pressure
Fiat money	Institutional credibility and policy discipline	Flexible and operationally scalable	Discretion, policy inconsistency, and dependence on trust
Bitcoin	Code-based supply rule and proof-of-work mining	Transparent technical scarcity and real mining cost	Indirect energy link; value still depends on market demand and coordination
Energy-linked digital finance	Reliable data plus rule-bound issuance, pricing, settlement, and governance limits	Potential link between digital claims and real production	Must verify data, price risk, protect settlement, and limit governance discretion

Table 2.1 separates technical scarcity from financial credibility. Technical scarcity limits creation under a rule. Financial credibility requires users to understand why the limit matters, what obligation follows from the claim, and whether that obligation can be enforced. The proposed framework asks whether energy can connect those two forms of discipline when evidence, pricing, settlement, and governance are explicit.

2.3 Gold, Bretton Woods, and Physical Monetary Constraint

Gold-backed money illustrates the strength and incompleteness of a physical constraint. Convertibility tied monetary claims to a scarce asset, and redemption pressure could expose over-issuance (Bordo, 1993). The constraint was therefore not gold alone; it was gold combined with a settlement promise.

That promise depended on institutions. Users relied on banks, governments, vaults, auditors, and reserve management rather than inspecting the gold directly. Physical scarcity did not remove custody, verification, geographic concentration, or crisis liquidity risk.

Bretton Woods made the dependence on settlement capacity explicit. Other currencies were tied to the U.S. dollar, and official foreign holders could convert dollars into gold at $35 per ounce. The system depended not only on the gold stock but also on confidence in U.S. reserve management, policy discipline, and political commitment (Eichengreen, 1992; Federal Reserve History, 2013).

As foreign dollar holdings expanded, the reserve issuer faced the Triffin tension: supplying global liquidity increased the stock of claims that might later seek conversion. The physical asset remained scarce, but the relationship between outstanding claims and credible settlement capacity became progressively harder to sustain.

By 1971, redemption pressure and policy conflict led the United States to suspend dollar-gold convertibility (Federal Reserve History, 2013; U.S. Department of State, n.d.).

The decisive failure was not a disappearance of gold. It was the breakdown of the rule connecting claims, reserves, and settlement.

The lesson for energy-linked finance is direct: a real asset or measured output does not constitute backing unless claim quantity and settlement obligations remain aligned with what the system can deliver.

Energy also differs materially from gold. It is time- and location-dependent, often non-storable in the economically relevant form, and inseparable from infrastructure and contractual delivery. An energy-linked system therefore cannot import gold convertibility by analogy; it must define production evidence, valuation, settlement, and governance separately.

2.4 Fiat Money and Institutional Credibility

Fiat money demonstrates a different route to credibility. Its support comes from law, taxation, payment infrastructure, central-bank authority, and broad acceptance rather than a fixed physical conversion promise.

Institutional flexibility permits liquidity management, lender-of-last-resort action, and crisis response. The same flexibility creates a discipline problem: users must believe that monetary discretion will remain consistent with the institution's mandate (Barro & Gordon, 1983; Kydland & Prescott, 1977).

Credibility is maintained through repeated performance, communication, legal authority, and anchored expectations (Federal Reserve Bank of St. Louis, 2010). It is therefore an institutional equilibrium, not a property of the monetary unit alone.

This distinction limits what a smart contract can borrow from sovereign money. Code can execute a transfer or supply rule, but it cannot create tax authority, deposit insurance, crisis liquidity, legal finality, or public accountability.

Most token issuers lack that institutional depth. An energy-linked claim therefore needs a narrower substitute that is explicit enough to inspect: accepted evidence, rule-bound issuance, visible risk treatment, protected settlement, and constrained authority to change the system.

The comparison does not imply that energy can replace fiat institutions. It clarifies the smaller claim tested here: whether energy evidence can discipline a defined financial obligation when institutional protections are limited.

2.5 Bitcoin, Protocol Scarcity, and Proof-of-Work

Bitcoin introduces a third form of discipline. Its issuance schedule is protocol-defined, while proof-of-work makes transaction-history revision costly (Nakamoto, 2008). It is neither redeemable like gold nor administered like fiat; its scarcity rests on code, distributed coordination, and resource-intensive mining.

Proof-of-work gives digital scarcity a real expenditure base. Miners consume electricity and hardware to compete for rewards, making Bitcoin costly to produce and secure. Yet costly production is not redeemable backing: holders receive no right to the electricity used in mining (de Vries, 2018).

Bitcoin is therefore a useful boundary case. A distinctive mining-expenditure effect must be separated from crypto-specific return dynamics, price persistence, ratio construction, and wider market behaviour (Granger & Newbold, 1974; Kristoufek, 2015; Kronmal, 1993; Liu & Tsyvinski, 2021).

A direct energy-linked claim would need to specify what Bitcoin leaves implicit: which evidence counts, when issuance is permitted, how uncertainty limits quantity, what must be settled, and who may change those rules.

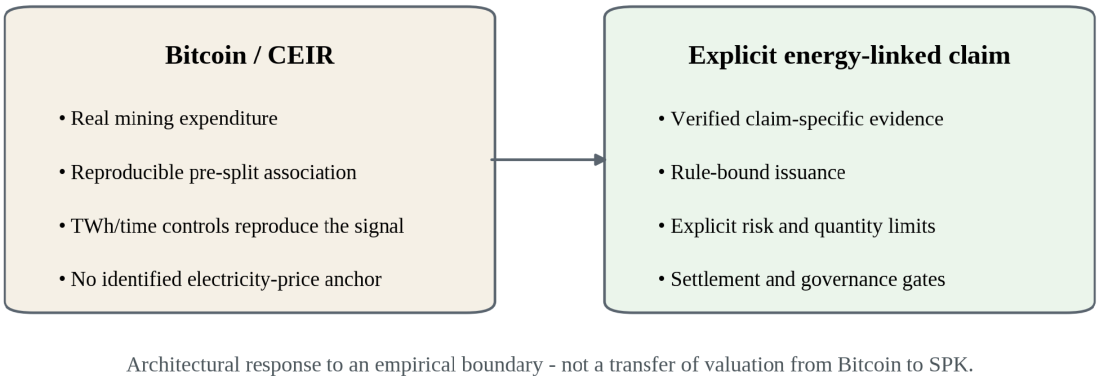

Figure 2.1. From passive mining expenditure to explicit claim constraints.

2.6 Bitcoin Energy Valuation and Cryptocurrency Returns

Production-cost research asks whether the resources required to mine Bitcoin help explain market value. Hayes (2019), for example, reports evidence consistent with a marginal production-cost view. Such results motivate an energy-cost test, but they do not establish a guaranteed value floor or a claim on the production input.

Mining cost changes with electricity prices, hardware efficiency, network difficulty, and miner behaviour, while market value responds to demand, liquidity, expectations, and coordination. A production-cost association must therefore be identified rather than inferred from the fact that mining is expensive.

The Cambridge Bitcoin Electricity Consumption Index provides model-based estimates of network power demand and electricity use (Cambridge Centre for Alternative Finance, n.d.-a, n.d.-c). The Mining Map separately describes the geographic distribution of hashrate (Cambridge Centre for Alternative Finance, n.d.-b). Neither source is direct metering of every miner, and any cumulative cost series inherits that uncertainty.

China's 2021 mining restrictions provide a plausible stress event because they changed the geographic distribution of hashrate. Geography matters only if the electricity-price path and the break specification identify a corresponding cost mechanism. A pre/post chart cannot supply that evidence on its own.

CEIR is therefore used as a falsifiable valuation-ratio test, not as a measure of intrinsic value. It compares market capitalisation with estimated cumulative mining electricity cost and asks whether the ratio predicts later returns. Replacing the denominator with cumulative TWh and cumulative time tests whether any apparent result is specific to electricity prices or merely a property of persistent series.

2.7 Renewable-Energy Finance, Data, and Risk

Renewable-energy finance changes the problem from costly expenditure to productive output. Solar and wind assets generate electricity with direct economic use. The financial value of that output nevertheless depends on weather, location, timing, grid access, storage, tariffs, curtailment, and settlement rules (Joskow, 2011; Ueckerdt et al., 2013).

Cost competitiveness does not eliminate those contingencies. Renewable generation can be economically attractive (Lazard, 2025) while still supporting a weak financial claim if production is poorly measured, risk is ignored, or settlement is undefined.

The financing problem is especially relevant in emerging and developing economies, where the International Energy Agency (2023) identifies capital constraints as a major obstacle. Digital infrastructure may lower some transaction costs, but it does not reduce physical, market, or settlement risk unless the claim itself becomes more credible.

Public data illustrate the evidence hierarchy. NASA POWER supplies satellite-based weather and solar-resource estimates, while PVWatts converts location and system assumptions into modelled production (NASA POWER, n.d.; National Laboratory of the Rockies, n.d.). These sources are useful for screening and scenario analysis, but they cannot prove that a specific site generated, exported, stored, or settled a given quantity of electricity (Dobos, 2014; Sengupta et al., 2024).

A credible claim must therefore distinguish resource potential, expected output, observed generation, contractual value, and settled delivery. Volatility, shortfall, basis risk, timing, and data quality must enter the financial design rather than remain hidden behind the energy label.

2.8 Pricing Energy-Linked Claims

Pricing provides the language for making those uncertainties explicit. Black and Scholes (1973) and Cox, Ross, and Rubinstein (1979) are not adopted because electricity behaves like an ordinary traded stock. They are useful because they force the analyst to declare the underlying proxy, volatility, horizon, discounting, and payoff.

Standard option assumptions fit electricity imperfectly. Power prices can spike, mean-revert, become negative, and respond to congestion and limited storability; renewable output also varies with weather and site conditions (Lucia & Schwartz, 2002). Option-style reasoning is therefore used as a transparent benchmark, not as a complete market model.

The Cox-Ross-Rubinstein tree is particularly useful in a cold-start setting because each step is reproducible and the convergence of the numerical result can be inspected. It supplies a disciplined scenario calculation when liquid options and implied volatility are unavailable.

Electricity-market research also shows why the benchmark must remain conditional. Non-storability changes arbitrage relationships, while market participants face operational and physical constraints in addition to price risk (Bessembinder & Lemmon, 2002; Deng & Oren, 2006; Lucia & Schwartz, 2002).

The relevance of pricing is operational rather than rhetorical. If uncertainty, location, data error, or settlement timing are ignored, the system may create a precise claim that exceeds what its evidence and capacity can support.

2.9 Smart Contracts, Oracles, Tokenisation, and Governance

Smart contracts can enforce issuance, transfer, redemption, and settlement rules, but execution quality is not economic quality. Automation can reduce some verification and enforcement costs while still faithfully applying a weak rule, accepting bad data, or creating an under-collateralised claim (Cong & He, 2019).

Tokenisation makes obligations programmable and can place multiple stages on a common settlement infrastructure (Bank for International Settlements, 2023). The token itself, however, does not establish what the claim represents, how the obligation is verified, or whether settlement capacity exists.

The oracle problem is therefore part of the financial design. A blockchain cannot observe whether electricity was generated, exported, curtailed, stored, or settled. External data bridges provide that information but introduce reliability, manipulation, and governance risk (Chainlink, 2025; Eskandari et al., 2021; Zhang et al., 2016).

Multiple sources can reduce dependence on one provider, but redundancy does not convert poor measurement into truth. A defective meter, weak reporting process, or modelled estimate remains defective or modelled when placed on-chain.

Governance creates the parallel institutional risk: an administrator can defeat the constraint without falsifying the data.

Energy does not limit a system if authorised parties can mint without accepted evidence, ignore redemption, or exempt preferred claims from the declared policy.

The same problem arises when pricing rules, oracle sources, or policy thresholds can be changed without delay, separation of roles, or an audit trail (Eskandari et al., 2021; Kiayias & Lazos, 2022).

Smart contracts can therefore express only part of the architecture. The system must first determine whether a claim should exist and how large it may be; execution and settlement follow later. Real deployment would still require legal agreements, audits, live data infrastructure, reserves, dispute procedures, cybersecurity, and regulatory analysis.

2.9.1 Relation to Stablecoin Design (Comparison, Not Identity)

The framework is not a stablecoin design. It does not promise a fixed dollar value, identify energy as a self-stabilising reserve, or claim that a physical reference can maintain a peg.

Stablecoin design begins with a liability and peg: who holds the reserves, how redemption works, how liquidity is supplied under stress, and how attestations are published. This thesis asks an earlier question: can verified energy evidence restrict the creation and settlement of a defined claim? U.S. dollars appear only as a valuation reference, not as promised parity (Bank for International Settlements, 2023).

A full stablecoin would require reserve policy, legal classification, liquidity management, and scalable peg operations. The Sepolia prototype tests only selected constraint and accounting rules and establishes no dollar parity.

2.10 Research Gap and Five Constraints

Across these literatures, credibility emerges from a chain rather than a single asset or technology. Gold requires settlement discipline, fiat requires institutional commitment, Bitcoin provides protocol scarcity without redemption, renewable energy provides uncertain production, pricing makes risk visible, and smart contracts enforce only the rule they are given.

The unresolved gap is the connection among those stages. Bitcoin research studies mining expenditure without direct energy claims. Renewable-energy finance studies production without digital issuance rules. Pricing research studies uncertain payoffs without specifying how the result should limit claim creation. Smart-contract research studies enforcement without deciding whether the underlying claim is economically defensible.

The thesis converts that gap into five jointly necessary constraints:

1. Evidence must match the claim. Satellite irradiance, modelled output, meter readings, exported electricity, and final settlement are different evidentiary states.

2. Claim creation must follow defined rules. If issuance can proceed without accepted evidence, energy is not limiting the system.

3. Uncertainty must be priced or bounded. Volatility, shortfall, basis risk, timing, and data error must change a financial decision rather than appear only in disclosure.

4. Settlement obligations must be defined. The system must state what is owed, what capacity supports it, and how collateral, reserves, shortfall, dispute, or redemption are handled.

5. Rule changes must be constrained. Versioned policy, separated authority, delays, audit trails, and emergency procedures determine whether the other limits can be trusted.

Energy remains an imperfect constraint: it is time- and location-dependent, measurement can fail, and financial value also depends on demand, liquidity, law, and trust. The thesis therefore does not treat energy as money. It tests whether energy evidence can become one enforceable limit within a broader financial architecture.

Chapter 3 - Empirical Evidence from Bitcoin Energy Costs

3.1 Research Question and Identification Problem

Chapter 3 tests the shortcut that motivates the rest of the thesis: whether real mining expenditure already anchors digital value. Bitcoin is the appropriate case because proof-of-work converts electricity and computation into a measurable cumulative cost.

The identification question is stricter than whether CEIR predicts later returns. If market value relative to cumulative mining cost matters, does the signal come from electricity prices, or can it be reproduced by cumulative energy use, elapsed time, price persistence, or an arbitrary starting stock?

A predictive coefficient cannot identify an energy mechanism by itself. The relationship must differ from negative-control ratios, survive changes to the initial cumulative-cost stock, remain meaningful in changes rather than only levels, and be supported by an appropriate structural-break test.

3.2 Why Bitcoin Is a Useful Case

Bitcoin separates expenditure from claim. Mining consumes real electricity, but holders possess no redeemable right to that electricity. This makes Bitcoin a useful test of whether costly production nevertheless leaves a distinctive valuation trace.

Three features make the case informative.

First, mining produces a cumulative expenditure series that can be compared with market capitalisation. CEIR makes that comparison, but both the numerator and denominator are highly persistent. Negative controls are therefore necessary to determine whether the result contains information specific to electricity cost.

Second, China's 2021 mining restrictions materially changed the geographic distribution of hashrate (Cambridge Centre for Alternative Finance, n.d.-b). The event provides a plausible candidate split, not proof that the cost mechanism changed.

Third, the regime interpretation is testable. Separate regressions and the classical Chow statistic describe the pre/post difference, while the joint robust Wald test evaluates whether the coefficients changed together with statistically reliable precision.

The ex ante hypothesis was that concentrated mining might create a more common production-cost reference and that later geographic dispersion could weaken it. The analysis treats that proposition as a hypothesis to be challenged, not a story to be confirmed.

Ethereum's transition from proof-of-work to proof-of-stake provides only contextual comparison (Ethereum.org, n.d.). Bitcoin remains the main case because proof-of-work operates throughout the sample.

3.3 Measuring the Energy-Valuation Relationship

The empirical test is organised around the Cumulative Energy Investment Ratio (CEIR).

The working definition is:

$$CEIR_t = \frac{Market\ Capitalisation_t}{Cumulative\ Energy\ Cost_t} \tag{3.1}$$

The main explanatory variable is $\log(CEIR_t)$. A high value means that Bitcoin's market capitalisation is large relative to estimated cumulative mining electricity cost. The expected coefficient is negative: an asset that appears expensive relative to cumulative cost should earn weaker subsequent returns. Because the denominator is cumulative and persistent, that interpretation is vulnerable to spurious regression and ratio construction (Granger & Newbold, 1974; Kronmal, 1993).

CEIR resembles a valuation ratio, but its construction creates the central identification problem. Market capitalisation carries the market-price signal in the numerator, while the denominator accumulates slowly over time. A negative coefficient may therefore describe persistence rather than an energy-cost channel.

An energy-specific result should outperform ratios based on cumulative TWh or cumulative days and remain stable when the starting cost stock, trend, and break specification change.

Tables 3.1-3.5 document the panel and its construction. Table 3.1 also records a decisive data limitation: the stored electricity-price path does not contain the intended continuous geography-weighted series. Table 3.6 places the reproduced association beside the tests that determine what it can and cannot mean.

3.3.1 Data sources and sample

Table 3.1 distinguishes external inputs from local and derived artifacts, while Table 3.2 defines the sample. The regression sample is smaller than the full panel because forward returns, rolling volatility, sentiment, and controls require complete observations.

Table 3.1. Data sources for the empirical panel.

Series	Source / construction	Frequency	Role
Bitcoin price and derived market capitalisation	Local BTC price export; Open used as Price; market capitalisation derived with an approximate BTC supply curve	Daily	Return outcome; CEIR numerator
Mining electricity	Cambridge CBECI best-guess annualised TWh export	Daily model estimate	Energy-cost base
Legacy electricity-price input	Canonical electricity_price column; intended geography-weighted merge failed	Daily; near-constant	Legacy cost denominator
Mining geography / candidate split	Cambridge Mining Map monthly shares, manually structured; daily split researcher-defined	Monthly / event split	Geography context
Fear & Greed Index	Alternative.me /fng/ historical API values	Daily	Composite sentiment control

Data and code note. Canonical input: bitcoin_ceir_analysis_ready.csv, derived from bitcoin_ceir_complete.csv. Inference script: thesis_package/ceir_regression.py. The repository contains no single end-to-end panel builder; --refresh-panel updates only the forward-return field.

The source audit corrected several earlier labels. The canonical panel does not use observed CoinGecko market capitalisation. It uses a local daily price export (btc_ds_parsed.csv) and an approximate Bitcoin supply curve to derive Market_Cap.

The upstream vendor and request metadata for that price export are not preserved, which limits exact source replay.

CBECI is a model-based best-guess estimate rather than direct miner metering (Cambridge Centre for Alternative Finance, n.d.-a). The Mining Map contains monthly pool-geolocation shares and does not designate 20 June as an official daily break (Cambridge Centre for Alternative Finance, n.d.-b). Fear & Greed is a third-party composite from Alternative.me's historical API (Alternative.me, n.d.). Appendix B.6 consolidates these source boundaries.

Table 3.2. Sample period and observation counts.

Sample	Start	End	N (days)	Pre-split	Post-split
Full analysis panel	2019-01-01	2025-05-28	2340	901	1439
Regression sample (controls complete)	2019-01-30	2025-04-28	2280	872	1408

Candidate split: 2021-06-20, defined by the researcher to represent the broader June 2021 mining crackdown. Cambridge does not provide this exact daily break.

Table 3.3. Variable definitions.

Symbol	Definition	Units
$CEIR_t$	Market Capitalisation$_t$ / Cumulative Energy Cost$_t$	Ratio (x)
$\log(CEIR_t)$	Natural log of CEIR; 1% winsorized in preferred regression (log_ceir_w)	log points
$R_{t,t+30}$	$Price_{t+30}$ / $Price_t$ - 1	Proportion
vol30	Rolling 30-day std of daily returns	Proportion
fg	Standardised Fear & Greed Index	z-score
trend	Linear time index (0 … T)	Days
post_china_ban	1 if Date ≥ 2021-06-20 (researcher-defined candidate split)	Indicator

Table 3.4. Descriptive statistics (regression sample; N = 2,280).

Variable	Mean (full)	Mean (pre-split)	Mean (post-split)	Standard deviation (full)
Bitcoin price (USD)	34,881	16,666	46,161	25,236
Derived market capitalisation (USD)	673,488,636,070	308,430,925,588	899,575,513,584	500,162,791,854
CEIR	29.6516	30.3994	29.1885	14.7314
log(CEIR)	3.2819	3.2850	3.2800	0.4536
Daily return	2.05e-03	3.51e-03	1.14e-03	0.0341
30-day forward return	0.0644	0.1072	0.0379	0.2080
Fear & Greed Index (0-100)	49.5009	52.2867	47.7756	21.8552
30-day return volatility	0.0318	0.0373	0.0283	0.0124

Table 3.5. Correlation matrix (regression sample).

Variable	log(CEIR)	30d forward return	Daily return	30d volatility	Fear & Greed
log(CEIR)	1.000	-0.189	0.040	0.273	0.353
30d forward return	-0.189	1.000	0.021	0.015	0.134
Daily return	0.040	0.021	1.000	9.76e-03	0.211
30d volatility	0.273	0.015	9.76e-03	1.000	-0.141
Fear & Greed	0.353	0.134	0.211	-0.141	1.000

The table reports Pearson correlations for the regression sample. Because 30-day forward returns overlap from one day to the next, the observations are serially related. Formal inference uses HAC(30) standard errors in Table 3.6 rather than relying on the raw correlations.

3.4 Empirical Design

The empirical design separates prediction, regime change, and mechanism.

The first test asks whether CEIR predicts Bitcoin's next 30-day return. A higher CEIR indicates higher market value relative to cumulative energy cost, so the predicted sign on log(CEIR) is negative.

The baseline predictive regression can be written as:

$$R_{t,t+30} = \alpha + \beta log\left(CEIR_t\right) + \gamma' X_t + \varepsilon_t \quad \textbf{(3.2)}$$

$R_{t,t+30}$ is the forward 30-day Bitcoin return, and X_t contains trend, sentiment, and volatility where available. Because adjacent forward-return windows overlap, conventional standard errors would overstate precision. The preferred inference uses HAC(30) standard errors, with month clustering and differenced specifications as additional checks (Hodrick, 1992; Newey & West, 1987).

The second test asks whether the relationship changes around the researcher-defined split of 20 June 2021. Separate regressions and the classical Chow statistic describe the before-and-after difference. The joint robust Wald test is the preferred test of whether the coefficients change together with statistically reliable precision (Chow, 1960).

The third test asks whether the apparent effect is specific to energy cost. The analysis removes the initial cost stock and replaces cumulative cost with cumulative TWh and cumulative days. It also estimates differenced specifications, examines stationarity and cointegration, applies alternative error structures, and evaluates a simple trading rule (Dickey & Fuller, 1979; Engle & Granger, 1987; Granger & Newbold, 1974).

Together, these tests distinguish an energy-price mechanism from a persistent valuation ratio that would arise under several cumulative denominators.

3.5 Main Results

The level regression reproduces the attractive result: before the candidate split, a higher CEIR is associated with weaker subsequent returns. Table 3.6 reports that association beside the diagnostics needed to identify it.

The pre-split coefficient on winsorised log(CEIR) is negative and statistically significant. The post-split coefficient remains negative but is smaller and insignificant. Although the classical Chow test rejects equal coefficients, the joint Wald test has HAC $p = .133$ and does not reject joint stability. The coefficients differ descriptively; the preferred test does not establish a structural break.

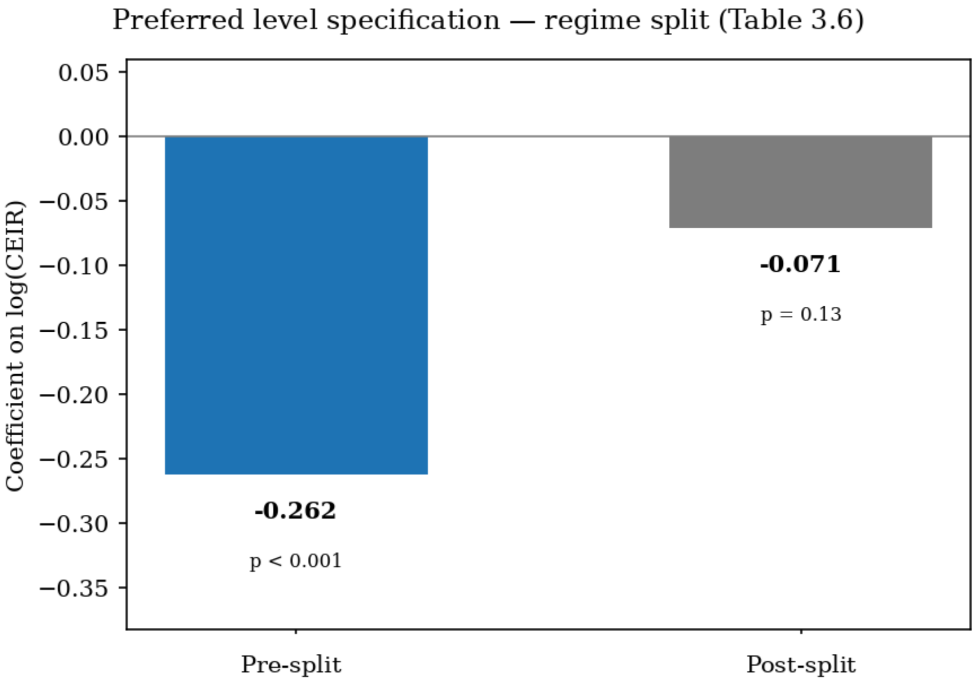

Figure 3.1. CEIR coefficients around the candidate split.

Table 3.4 and Figure 3.2 provide descriptive context. Bitcoin's mean price rises sharply after the split, while mean CEIR remains in a broadly similar range. The distributions overlap but shift, and the raw correlation between log(CEIR) and forward returns is negative. Because the return windows overlap, formal inference rests on HAC(30) standard errors rather than that correlation.

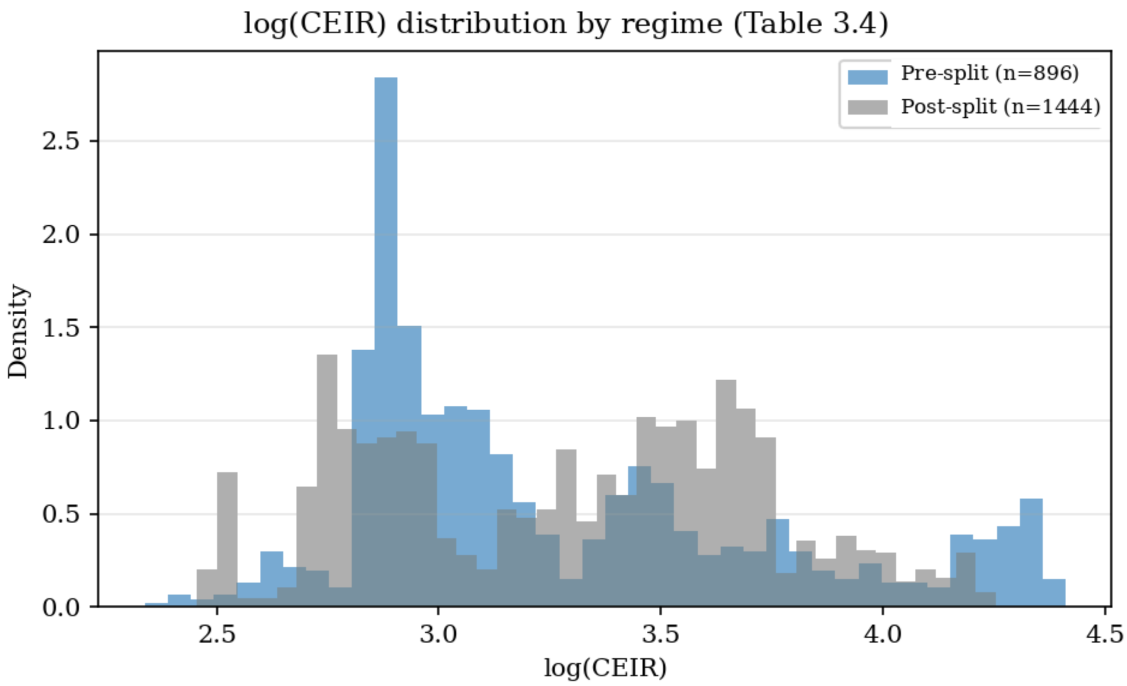

Figure 3.2. CEIR distribution around the candidate split.

Note: Figure 3.2 uses the full CEIR panel for visual distribution only (n = 896 pre-split; n = 1,444 post-split). Tables 3.4 and 3.6 use the regression-ready sample with complete controls (N = 872 pre-split; N = 1,408 post-split).

The original specification is therefore reproducible: a higher pre-split CEIR is associated with weaker returns over the next 30 days. Reproduction, however, is not identification. Placing cumulative mining cost in the denominator does not show that electricity prices caused the relationship.

The negative controls point to a persistent valuation signal rather than an energy-price anchor. Market capitalisation divided by cumulative TWh produces almost the same coefficient, while market capitalisation divided by cumulative days produces a similar and slightly stronger result. The pre/post pattern cannot be identified as a geography-sensitive electricity-cost mechanism.

Table 3.6. Legacy association and final identification diagnostics.

Test	N	Estimate	Robust p	Reading
Pre-split CEIR	872	β = -0.2623	.00052	Association
Post-split CEIR	1,408	β = -0.0708	.133	Weak
Joint Wald	2,280	—	.133	No robust break
Zero 2018 seed	872	β = -0.1037	.099	Seed-sensitive
TWh / days controls	872	β = -0.2618 / -0.2811	.00052 / .00027	Not energy-specific

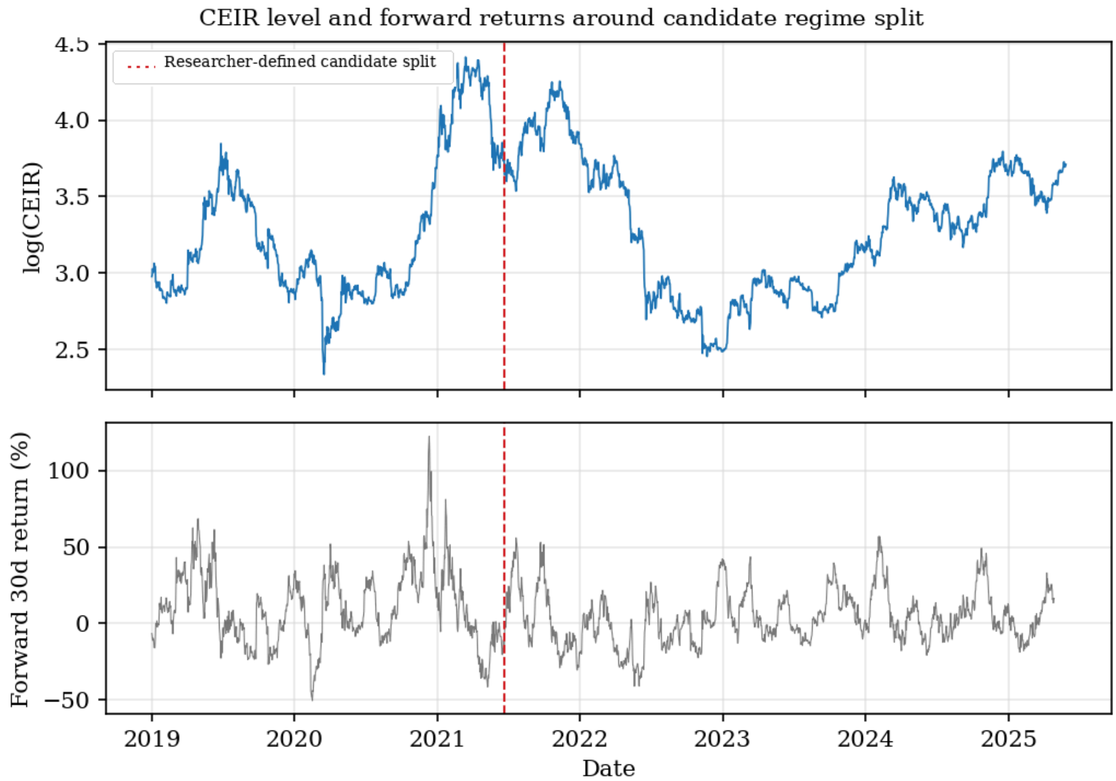

Figure 3.3. CEIR timeline and forward-return panel.

Figure 3.3 locates the candidate split on the CEIR and forward-return timeline. It is descriptive; the formal regime evidence remains the set of break tests reported in Table 3.6.

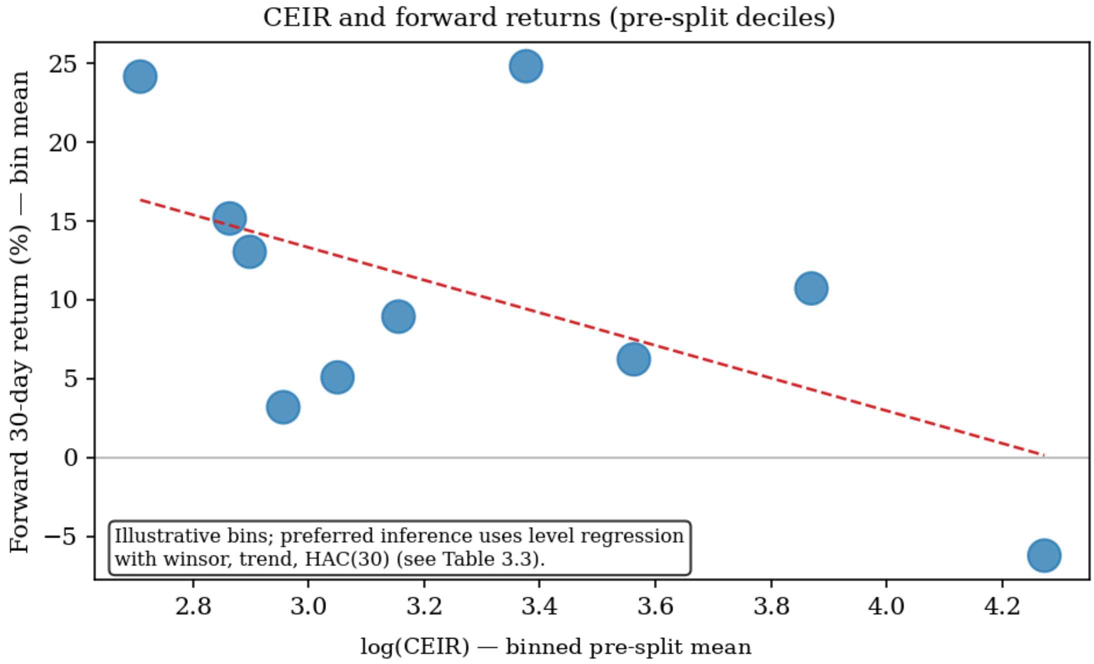

Figure 3.4. Illustrative pre-split decile means; not the preferred regression.

3.6 Robustness and Negative Results

The robustness analysis does not merely qualify the level result; it changes its interpretation.

Differencing removes the apparent predictive effect. Short-run changes in CEIR do not reliably predict later returns, which is consistent with the level coefficient being driven by persistent series rather than an energy-cost signal (Granger & Newbold, 1974).

The ratio also fails a practical relevance check. Over the same sample, the CEIR strategy earns about +176%, compared with about +2771% for buy-and-hold; the corresponding Sharpe ratios are 0.72 and 1.13. The backtest provides no evidence that the signal is commercially useful (Lo, 2002).

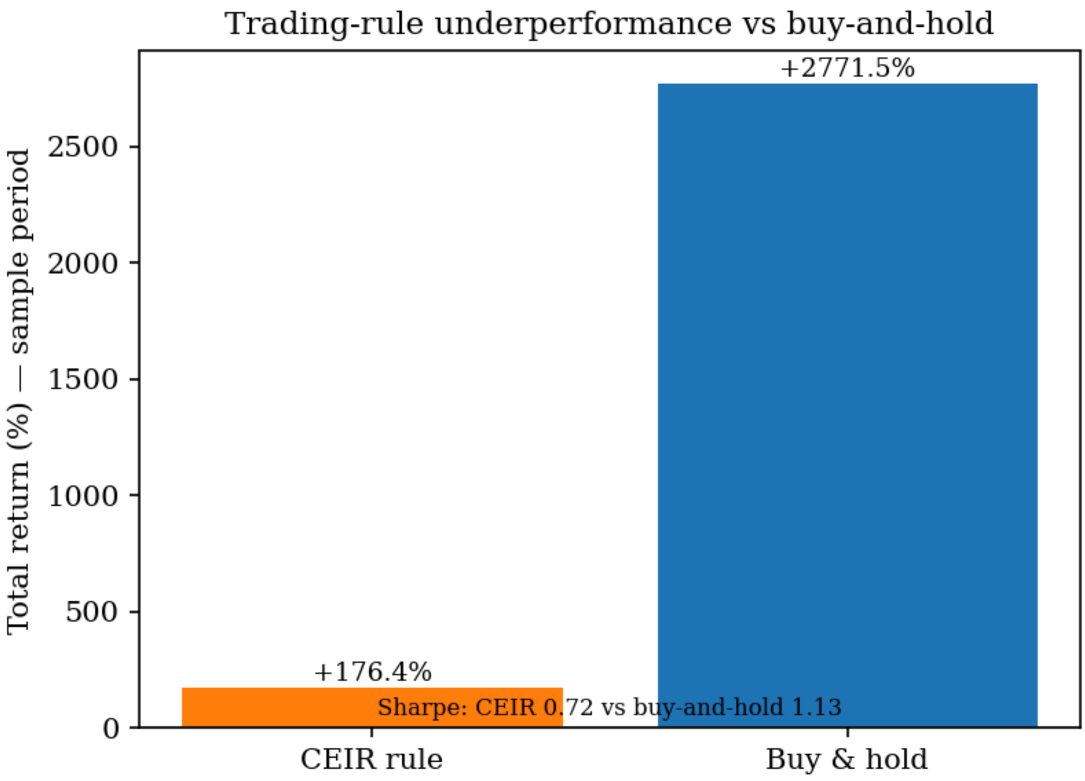

Figure 3.5. Trading-rule negative result.

The data audit then identifies two structural problems in the denominator. First, the stored electricity-price series is almost constant at about $0.076/kWh because the intended monthly geography-weighted merge failed, apart from a few month-start overrides.

Second, the 2019 window begins with approximately $3.30 billion of cumulative 2018 cost. This amount is the pre-sample CBECI × stored-tariff cost carried into the analysis, not an unexplained external seed. Setting that stock to zero weakens the pre-split coefficient from about -0.262 to -0.104 and raises the HAC p-value to about .099. The apparent anchor is materially sensitive to a pre-sample accounting choice.

The negative controls are even more decisive. CEIR and market capitalisation divided by cumulative TWh have a log-ratio correlation of .999989 and nearly identical pre-split coefficients. A ratio based on cumulative days is also significant with a similar coefficient. Electricity prices are not supplying unique information.

Time-series diagnostics reinforce that conclusion. ADF tests do not reject unit roots in log(CEIR), log market capitalisation, or log price. Engle-Granger tests also do not support a stable long-run equilibrium between market capitalisation and either cumulative cost or cumulative TWh (Dickey & Fuller, 1979; Engle & Granger, 1987).

The joint robust break test remains insignificant, and regressions that include both price and CEIR are unstable because the variables are mechanically linked. The analysis also covers only one proof-of-work asset. Taken together, the evidence supports a bounded conclusion: passive mining-cost ratios do not cleanly identify an energy-value anchor.

3.7 Interpretation and Implications

The first implication is empirical. Bitcoin mining expenditure is real, but CEIR does not isolate its financial effect. Cumulative energy use and cumulative time reproduce the same broad relationship, so the coefficient cannot be interpreted as information unique to electricity prices.

The second implication concerns regime narratives. The pre- and post-split estimates differ, but the robust joint Wald test does not reject stability. Mining geography and cost dispersion remain plausible mechanisms; this panel does not identify them as causes.

The third implication is constructive. Passive expenditure neither determines price nor gives holders an energy claim, collateral rule, or settlement right. The negative result shifts the thesis from inferred backing to explicit constraints.

3.8 Empirical Conclusion

The evidence does not identify a clean energy-specific financial constraint. The pre-split CEIR association is reproducible, but it weakens when the starting cumulative-cost stock is removed.

It is almost indistinguishable from a cumulative-TWh ratio, resembles a cumulative-days ratio, is unsupported by the preferred robust break test, fails in differences, and produces no useful trading advantage.

The chapter therefore removes an unsupported shortcut: spending real energy does not automatically back a financial claim. Any credible energy link must be created through explicit rules for evidence, issuance, risk, settlement, and governance.

Chapter 4 - Pricing Renewable-Energy Risk

4.1 Pricing Problem and Scope

Chapter 3 rejects passive mining expenditure as a clean value anchor. Chapter 4 asks the constructive question that follows: how should a financial claim treat renewable-energy uncertainty when no liquid market supplies an observable price?

Renewable output varies with location, season, weather, equipment, and grid conditions. A claim that ignores those sources of variation can promise more than its underlying production relationship can support (Joskow, 2011; Ueckerdt et al., 2013).

The objective is not to discover a universal energy price. It is to make the assumptions behind an energy-linked payoff explicit enough to inspect, stress, and eventually connect to policy.

The chapter develops a cold-start scenario framework. Public energy data supply a production-risk proxy; a declared payoff is valued with binomial and Monte Carlo methods; and the resulting quantities are translated into margin and data-tolerance diagnostics.

The contribution is therefore a financial decision layer. Energy evidence becomes operational only when uncertainty changes collateral, admission, quantity, or settlement protection.

4.2 Why Pricing Comes Before Settlement

A physical reference is financially incomplete without a valuation rule and a treatment of the risks that can move the obligation.

For a contract linked to renewable output, the relevant questions are immediate. What is the reference value? How volatile is the underlying proxy? How much collateral absorbs stress, how much measurement error is tolerable, and do the same parameters remain defensible across locations?

These questions precede automation. A smart contract can execute an underpriced rule perfectly. If volatility, basis risk, or shortfall is understated, technical correctness can coexist with financial fragility.

Pricing is therefore used as a constraint input: it exposes the uncertainty that issuance and settlement rules must absorb.

4.3 The Underlying Risk

Renewable output and value vary for different reasons. Irradiance, weather, season, equipment, and grid conditions affect production; timing, location, market structure, and contract terms affect the value of a kilowatt-hour (Joskow, 2011; Lucia & Schwartz, 2002; Ueckerdt et al., 2013).

Solar is used because public resource and production models are accessible. NASA POWER and PVWatts support calibration and scenario comparison (NASA POWER, n.d.; National Laboratory of the Rockies, n.d.). They do not replace meter or inverter records when a real settlement claim is made (Dobos, 2014; Sengupta et al., 2024).

4.3.1 Location inputs and volatility calibration

Table 4.1 records the declared inputs used by the pricing script. S_0 and r are scenario parameters rather than preserved observations, because the audited files do not contain a complete dated source ledger for all five locations.

Taiwan's volatility is the one directly supported calibration. The archived diagnostic applies a four-day rolling mean and a 1% absolute-log-return trim, producing annualised σ = 189.5% from 2,166 transformed returns (Jarque-Bera p = .349). The other location volatilities remain declared scenarios because their raw NASA requests and calibration diagnostics are not preserved.

Table 4.1. Location parameters and volatility calibration inputs.

Location	Declared S_0 proxy ($/kWh)	σ (annual, %)	Declared r input	Input status
Taiwan	0.0525	189%	2.5%	σ calibrated in archived diagnostic
Saudi Arabia	0.0550	172%	2.5%	Declared scenario input
Arizona, USA	0.0580	165%	4.5%	Declared scenario input
Brazil	0.0950	198%	13.5%	Declared scenario input
Germany	0.0250	45%	3.5%	Declared scenario input

All locations use T = .25 years and an at-the-money convention in which $K = S_0$. The calculations are implemented in options_pricing.py.

The archived calibration records coordinates, a 2019-2024 window, a four-day rolling mean, and a 1% tail trim. Exact API replay is incomplete because the POWER parameter code and UTC versus Local Solar Time setting were not preserved. That provenance gap limits replication of the raw request, not the transparency of the archived transformation.

Table 4.2. Binomial tree convergence (Taiwan base case).

Steps (N)	Option Price ($/kWh)	Change from Previous
50	0.0191	—
100	0.0191	+0.247%
200	0.0192	+0.124%
400	0.0192	+0.062%
800	0.0192	+0.031%
1200	0.0192	+0.010%

The binomial value stabilises rapidly; N = 400 is used as the preferred balance between convergence and computational cost.

Table 4.3. Margin stress grid (selected S_0 and σ).

S_0	σ	VaR_{99}	Initial margin (1.5×)
$0.0420	142%	$0.1776	$0.2665
$0.0420	189%	$0.3378	$0.5066
$0.0420	236%	$0.6146	$0.9219
$0.0525	142%	$0.2220	$0.3331
$0.0525	189%	$0.4222	$0.6333
$0.0525	236%	$0.7682	$1.1524
$0.0630	142%	$0.2665	$0.3997
$0.0630	189%	$0.5066	$0.7599
$0.0630	236%	$0.9219	$1.3828

Initial margin is set to 1.5 × VaR_{99} of spot. The Taiwan base case uses S_0 = $0.0525/kWh and σ = 189%. The full stress grid appears in margin_stress_table.csv.

4.4 Model Setup

The model treats a declared dollar-per-kilowatt-hour proxy as the underlying variable in a short-horizon call-style payoff. Its inputs are S_0, strike K, volatility σ, discount rate r, and horizon T. Because the irradiance/output proxy is neither freely traded nor directly hedgeable, the output is a conditional risk benchmark rather than an arbitrage-free market price (Bessembinder & Lemmon, 2002; Lucia & Schwartz, 2002).

Taiwan provides the primary numerical example because its volatility transformation is preserved. The cross-location cases show how declared inputs change the result; they do not claim equal evidentiary strength across sites.

Geometric Brownian motion is used as a transparent short-horizon approximation. It does not reproduce the jumps, seasonality, negative prices, or mean reversion observed in power markets. Its role is to make the payoff and sensitivity calculations explicit, not to close those modelling questions.

Table 4.4. Taiwan base-case inputs and pricing output.

Parameter	Value
Declared underlying proxy S_0	$0.0525/kWh
Strike/reference cost K	$0.0525/kWh
Horizon T	0.25 years
Declared discount-rate input r	2.5%
Volatility σ	189%
Binomial call value	$0.01917/kWh
Monte Carlo call value	$0.01957/kWh
Method gap	About +2.1% Monte Carlo vs binomial

4.5 Numerical Pricing Methods

The payoff is valued with a Cox-Ross-Rubinstein tree and Monte Carlo simulation (Boyle, 1977; Cox, Ross, & Rubinstein, 1979). Agreement between the two methods is an implementation cross-check under common assumptions; it is not validation of the assumptions themselves.

For Taiwan, the binomial value is approximately \$0.01917/kWh and the Monte Carlo value is \$0.01957/kWh, a gap of about 2.1%. Across locations, the high-volatility Taiwan and Brazil scenarios produce larger option values, while Germany produces a smaller value because both S_0 and σ are lower. Older fixed-strike runs are retained only as non-canonical robustness artifacts.

The convergence exercise shows that the Taiwan binomial estimate changes only marginally beyond roughly 400 steps. The cross-method and convergence checks establish numerical consistency within the declared model.

Table 4.5. Cross-location at-the-money option values.

Location	Declared S_0 proxy (\$/kWh)	σ (annual, %)	Binomial call (\$/kWh)	Monte Carlo call (\$/kWh)	Difference (MC vs B)
Germany	0.0250	45%	0.00234	0.00236	0.9%
Taiwan	0.0525	189%	0.01917	0.01957	2.1%
Saudi Arabia	0.0550	172%	0.01841	0.01876	1.9%
Arizona, USA	0.0580	165%	0.01877	0.01911	1.8%
Brazil	0.0950	198%	0.03702	0.03781	2.1%

For each location, the strike is set equal to its own spot proxy: $K = S_0$. Source: cross_location_pricing.csv.

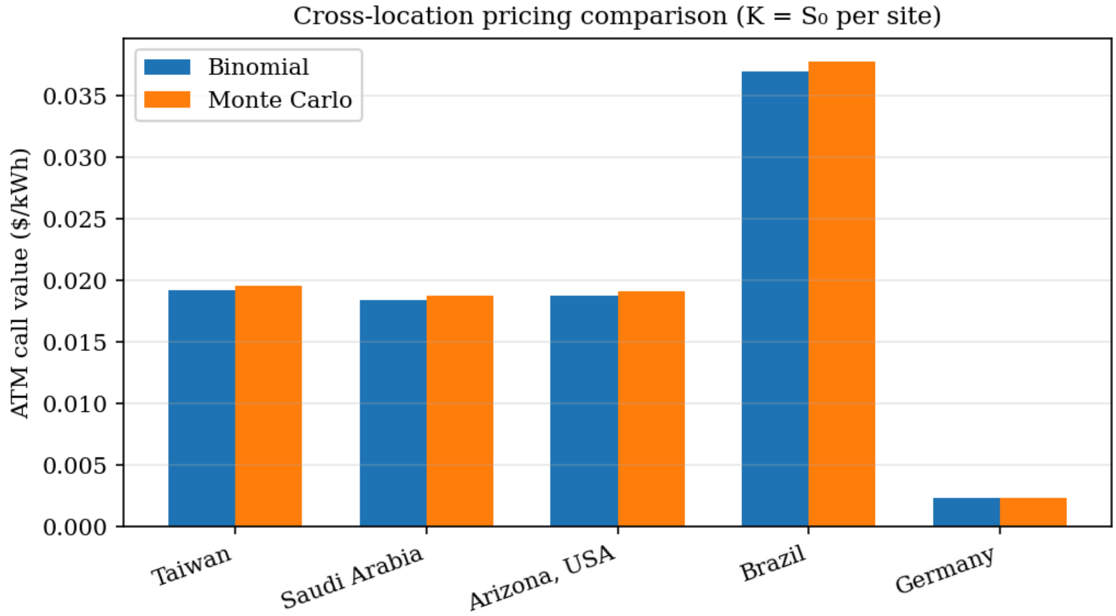

Figure 4.1. Cross-location at-the-money option comparison.

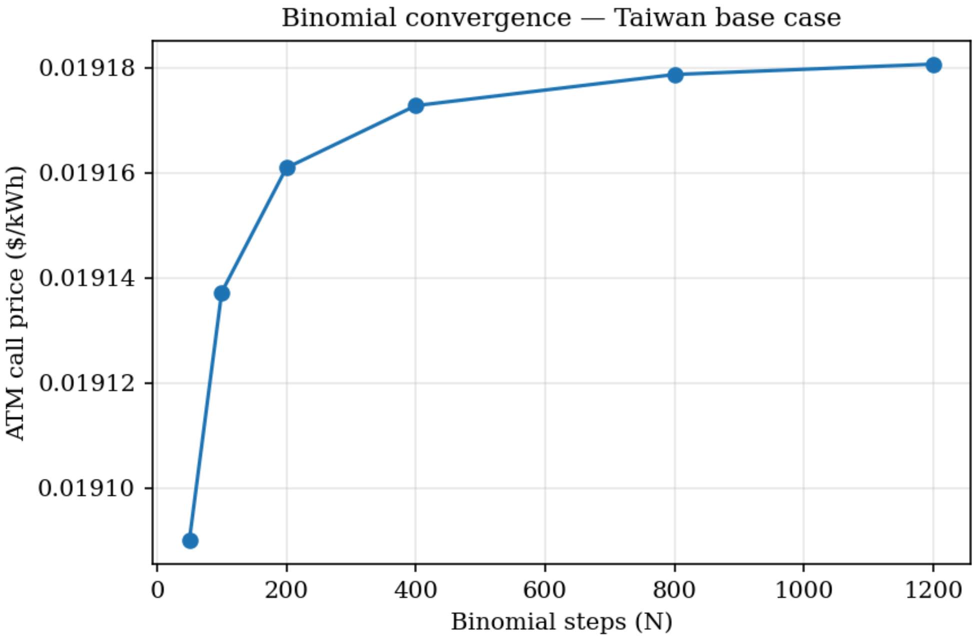

Figure 4.2. Binomial convergence for the Taiwan base case.

4.6 Cross-Location Results

The cross-location comparison demonstrates sensitivity, not ranking. Given the declared inputs, higher S_0 and σ produce larger option values and larger stress margins. Brazil and Taiwan therefore generate larger model outputs than Germany, but the result cannot be read as independent evidence that one location is economically riskier.

The useful output is a location-labelled scenario set whose assumptions can be inspected, changed, and rerun. That transparency is itself a constraint: a contract cannot hide a site-specific risk decision behind the generic description 'energy-backed'.

4.7 Collars, Oracle Tolerance, and Margin

The collar exercise shows how payoff design changes exposure. In the lognormal benchmark, equal percentage distances around spot need not create symmetric option values because the strikes are not equally distant in log space. The resulting net credit grows with σ; it is a model property, not a discovered market threshold.

The oracle-tolerance exercise asks how much measurement error can be absorbed before the modelled hedge falls below a declared variance-reduction target (Table 4.6; Figure 4.3). Under the common rule, the higher-volatility scenarios permit larger errors mechanically. The calculation does not measure real oracle quality; it shows how a policy could make data tolerance explicit.

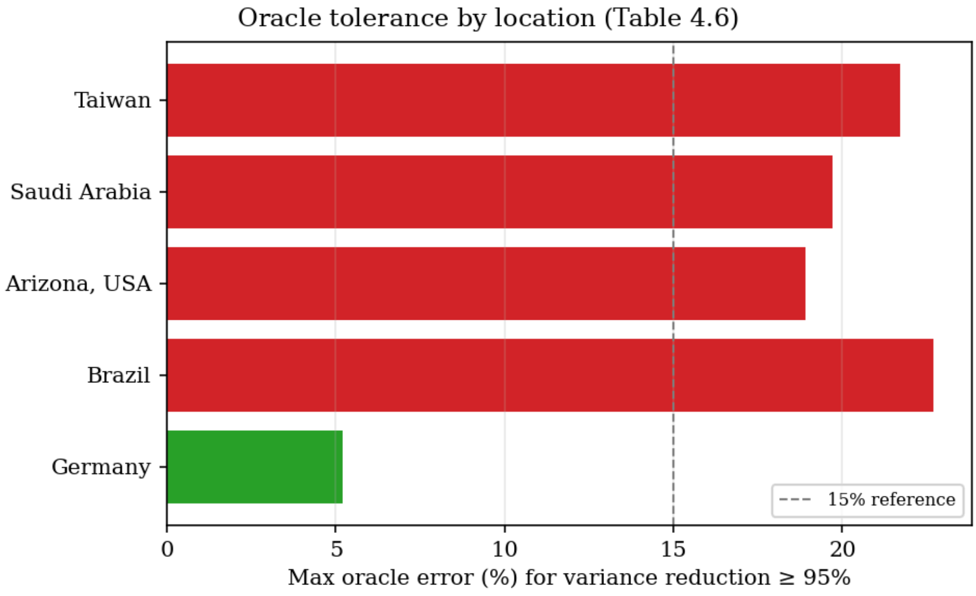

Figure 4.3. Oracle tolerance by location.

The margin stress grid gives the same logic a collateral interpretation. A declared 1.5 × VaR_{99} rule increases with S_0 and σ; for the Taiwan base case, it produces approximately $0.63/kWh of initial margin. This is a policy output under stated assumptions, not an observed market requirement.

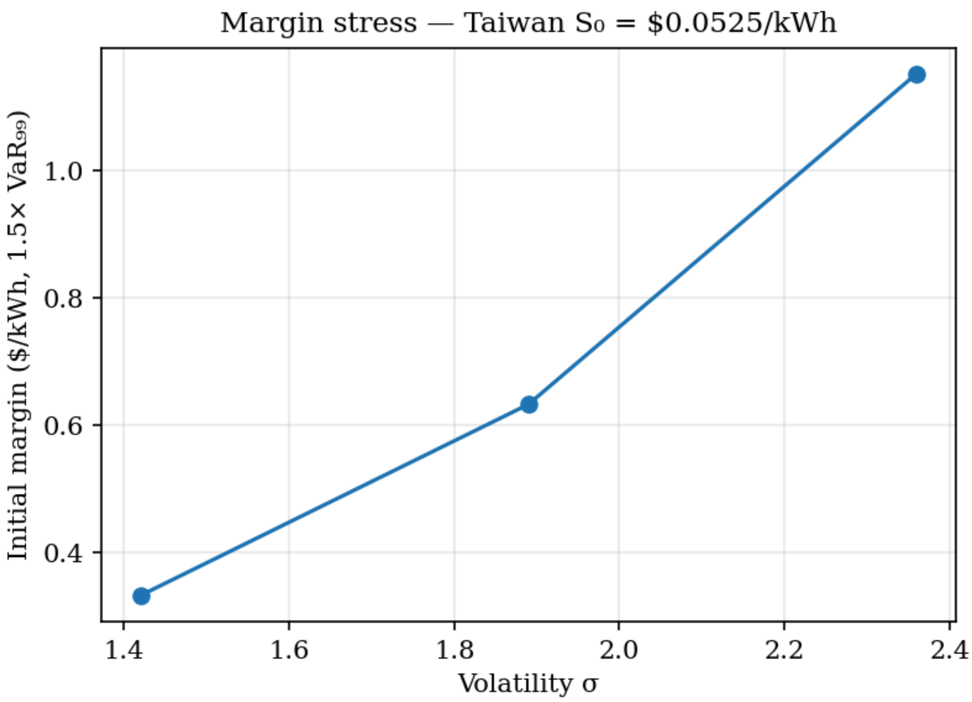

Figure 4.4. Margin stress for the Taiwan base case.

Table 4.6. Maximum oracle error for variance reduction ≥ 95%.

Location	Maximum Oracle Error for Variance Reduction ≥ 95%
Taiwan	21.7%
Saudi Arabia	19.7%
Arizona	18.9%
Brazil	22.7%
Germany	5.2%

4.8 What the Pricing Layer Proves and Does Not Prove

The chapter establishes three results. First, a renewable-output proxy can be translated into inspectable option, margin, and data-tolerance quantities under declared assumptions. Second, binomial and Monte Carlo methods agree closely enough to support the numerical implementation. Third, the outputs change materially with location-labelled inputs, making hidden parameter choices visible.

The evidentiary strength is uneven by design: Taiwan has a preserved volatility transformation, while the other locations remain declared scenarios because their calibration lineage is incomplete.

The methodological contribution is therefore not a market-price estimate. It is a reproducible path from modelled energy context to financial quantities that can be stress-tested. A later policy layer can map those quantities to margin, admission, quantity, or settlement protection (Boyle, 1977; Cox, Ross, & Rubinstein, 1979; Dobos, 2014; Sengupta et al., 2024).

One boundary governs the interpretation. The proxy is non-traded, the model is not arbitrage-free evidence, and public resource data are not physical settlement records. Liquidity, basis, contract enforceability, counterparty risk, demand, and delivery therefore remain outside the result (Bessembinder & Lemmon, 2002; Joskow, 2011; Lucia & Schwartz, 2002; Ueckerdt et al., 2013).

A calculated risk number becomes a constraint only when policy assigns it an operational consequence. Until it changes allowed quantity, required margin, admission, or settlement protection, it remains analysis rather than governance.

The current V2 case pack does not yet use the option benchmark as a validated volatility-based quantity ceiling. Its tested limits come from evidence-backed capacity, provenance policy, model-based resource context, and an absolute policy cap. Applying the pricing output would require a pre-declared conversion into compatible claim units.

4.9 Pricing Conclusion and Policy Boundary

Pricing is necessary but not self-executing. The chapter makes risk visible; Chapter 5 asks how evidence, risk, quantity, settlement, and governance can be joined into a rule-bound claim process.

Chapter 5 - Constraints Framework and Proof-of-Concept Implementation

5.1 From Financial Weakness to Explicit Constraints

The empirical chapter removes passive expenditure as a credible anchor, and the pricing chapter shows that a risk estimate has no force until policy uses it. The remaining problem is institutional: how can a system prevent claims from exceeding their evidence, permitted quantity, or settlement capacity?

The architecture answers that problem through five linked constraints. Each governs a different failure margin: evidence quality, issuance authority, financial risk, settlement capacity, and rule-changing power.

A working smart contract is not sufficient. Software can execute a financially weak rule exactly as written. The claim becomes credible only when the system can explain why it was admitted, what limited its size, what is owed later, and which policy authorised the decision.

Two proof-of-concept layers test those stages. SPK v1 on Sepolia examines selected rules after evidence has been accepted: attestation, issuance, payment, redemption, and settlement accounting. The V2 case workbench examines the prior decision: evidence and context are recorded, policy is applied, admission and quantity are determined, and later settlement is evaluated separately. Together they provide feasibility evidence, not a production financial system.

5.2 The Constraints Framework

The five constraints can be expressed as five questions that every energy-linked claim must answer.

First, does the evidence support the claim being made? Resource potential, modelled output, meter readings, exported electricity, and settled delivery are not interchangeable. The evidence threshold must rise with the strength of the financial obligation.

Second, is claim creation rule-bound? Accepted evidence must lead to a defined issuance decision rather than merely inform an administrator who retains unrestricted minting power.

Third, how does uncertainty constrain the claim? Volatility, shortfall, oracle error, and basis risk must change a compatible quantity, margin, or settlement safeguard rather than appear only as disclosure.

Fourth, what is owed at settlement and what happens when capacity is insufficient? A credible system must distinguish full performance, partial coverage, shortfall, dispute, and redemption accounting.

Fifth, who can alter the preceding rules? Versioned policy, separated authority, delays, and audit trails determine whether the other constraints bind in practice.

The constraints are jointly necessary. Reliable data without settlement protection still supports a weak claim; precise pricing of the wrong evidence produces a precise error; and rule-bound issuance is cosmetic if governance can exempt itself later. Table 5.1 summarises the architecture.

Table 5.1. Five constraints for credible energy-linked claims.

Constraint	Purpose	Failure if Missing
Reliable energy data	Defines what energy evidence the system accepts	The system may price or mint against false or weak claims
Rule-bound issuance	Limits token or contract creation to accepted evidence	The issuer regains discretionary creation power
Explicit pricing and risk controls	Accounts for volatility, basis risk, oracle error, and shortfall risk	Claims become underpriced or under-collateralised
Protected settlement and redemption accounting	Defines what is owed and what happens during fulfilment, shortfall, or dispute	Users hold claims without credible resolution rules
Limited governance	Restricts discretionary parameter changes and role abuse	The system can override its own constraints

5.3 Constraint 1: Reliable Energy Data

The first constraint is evidentiary separation. Resource potential, modelled output, actual generation, surplus export, and tariff value describe different states. NASA POWER and PVWatts provide model baselines and anomaly context, while meter or inverter records form the stronger path for an operational claim (Dobos, 2014; Sengupta et al., 2024).

The attested mint path illustrates how evidence can become a checkable input. Meter-style readings are submitted, accepted readings produce a fixed source hash, and a signed attestation binds the surplus amount to related metadata. Minting is permitted only after those checks pass.

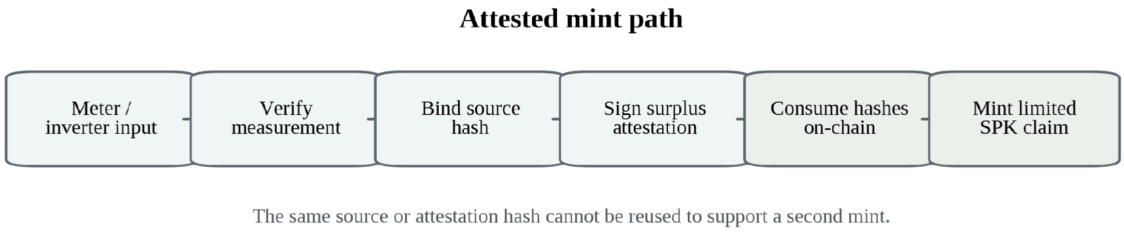

Figure 5.1. Data-to-mint pipeline: source and attestation hashes consumed on-chain.

The sample readings are not revenue-grade metering. What the proof establishes is narrower and still material: accepted and rejected evidence paths can be distinguished, bound to issuance, and protected against reuse.

5.4 Constraint 2: Rule-Bound Issuance

Rule-bound issuance requires more than an administrator's assertion that energy exists. The issuance event must be linked to evidence that the system has already accepted under a declared rule.

The surplus attestation binds the recipient, surplus kWh, measurement and validity windows, source hash, chain ID, and contract address. Source and attestation hashes are consumed after use, preventing the same evidence from supporting multiple mints. Issuance becomes a reproducible decision rather than discretionary authority.

The earlier Sepolia proof demonstrates the path with a signed bundle for 2,606.7 kWh of accepted surplus and 130.1697 SPK minted at a $0.05/kWh basis. The transaction proves that the rule can execute; it does not prove live operator metering, legal enforceability, or production readiness.

5.5 Constraint 3: Explicit Pricing and Risk Controls

The pricing constraint asks whether financial uncertainty has an operational effect. Chapter 4 makes location, volatility, margin, and data tolerance visible. The current contracts do not execute the full model on-chain, and the V2 policies do not present the benchmark as a validated quantity rule. Instead, the workbench defines where a declared risk calculation could enter as a compatible ceiling or margin requirement.

Financial parameters remain declared rules rather than market truths. The earlier proof used a $0.05/kWh conversion, while SPK v1 uses an energy-native default of 1 kWh -> 1 SPK. Neither creates a guaranteed price. The workbench compares only limits expressed in compatible units and evaluates settlement later against explicit capacity.

5.6 Constraint 4: Settlement and Redemption Accounting

Issuance is incomplete without a defined obligation. The currency-system contract records two later-stage processes: invoice settlement transfers SPK against a replay-protected invoice hash, while redemption accounting burns tokens into owed-kWh claims that can later be marked fulfilled, in shortfall, or disputed.

Those records do not guarantee physical delivery. Their contribution is accounting separation: minting, circulation, redemption, and resolution can be linked in one auditable path. Real deployment would still require named counterparties, legal terms, operator duties, and funded reserve or capacity arrangements.

5.7 Constraint 5: Governance Limits and Launch Gates

Governance determines whether the other four constraints remain credible. If one administrator can alter evidence rules, minting authority, pricing parameters, or settlement obligations at will, the system has recreated discretionary issuance under an energy label.

The design therefore uses separated roles, pausing, governance-delay patterns, and staged launch gates (Kiayias & Lazos, 2022). The public lab is open for research inspection;

closed-pilot and paid/mainnet stages remain blocked pending governed deployment, real operator data, stronger hardware provenance, economic and legal terms, audit, redemption policy, and production evidence.

Publishing the blockers is part of the method. The project treats readiness as a set of falsifiable gates rather than presenting testnet activity as finished money. Table 5.2 and Figure 5.2 show the separation among research, pilot, and product stages.

Table 5.2. Launch-gate stages and current status.

Stage	Current Status	Interpretation
Public lab	Open for proof, demo, and research use	Suitable for advisor, reviewer, and public research inspection
Closed testnet pilot	Blocked	Needs governed deployment, real operator data, stronger hardware provenance, and clear economic/support terms
Paid/mainnet product	Blocked	Needs audit, legal and commercial scope, redemption policy, production deployment, reserves, real counterparties, and demand

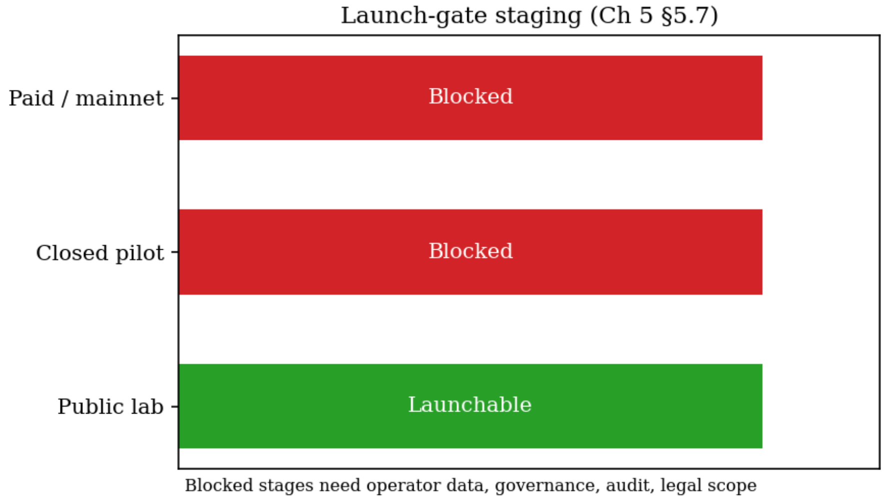

Figure 5.2. Launch-gate staging for research, pilot, and product phases.

5.8 Boundaries of the On-Chain Proof-of-Concept

The on-chain proof establishes a bounded capability set: evidence can pass through attestation and replay-protected minting; payments can use invoice-hash protection; and redemption outcomes can be recorded.

Because the transactions are public-testnet records, reviewers can inspect execution and distinguish research stages from later deployment claims.

The proof does not establish revenue-grade metering, enforceable delivery, reserve sufficiency, market demand, liquidity, external audit, legal classification, or production cybersecurity. Those are deployment conditions, not hidden assumptions of the result.

5.9 SPK v1 on Sepolia

The project first deployed a dollar-translated attested mint in May 2026. SPK v1 followed in June and retained the five-constraint logic while shifting the unit of account from a dollar translation toward energy-native circulation. (SolarPunk project artifacts, 2026).

The earlier proof converted accepted energy into dollars at $0.05/kWh. SPK v1 instead uses a default rule of 1 kWh -> 1 SPK, focuses on network payments, and leaves the dollar peg disabled. Appendix B records the canonical contracts and reviewed runtime snapshot.

SPK v1 contributes implementation evidence rather than a new valuation claim. Consumed hashes prevent repeated issuance, invoice hashes prevent payment replay, redemption is gated, and operator keys remain testnet-only.

The indexed record contains a 5 SPK wallet-initiated pilot transfer and a 180 SPK operator test action. Neither is treated as external commerce, and held or unallocated SPK is classified as testnet treasury or reserve rather than circulating float.

5.10 From Five Constraints to Deterministic Decision Stages

The central operational insight is that an energy-linked claim contains three distinct decisions: may the claim exist, how large may it be, and can it later be settled? The workbench preserves that sequence as case -> evidence and context -> policy -> admission -> quantity limits -> DecisionResult -> claim -> settlement -> receipt.

The software objects support that economic sequence. CaseManifest and ContextManifest preserve the declared case, evidence, and model context; ConstraintEvaluation records the rules applied; DecisionResult fixes admission and quantity; and DecisionReceipt preserves the later inspection record. A V2 claim must reference a verified DecisionResult. Evaluation time is stored in the receipt rather than the decision hash, so repeated evaluation of identical inputs preserves decision identity.

Admission returns PASS or BLOCK. Quantity is then limited by the lowest applicable ceiling among constraints expressed in the same claim unit. Settlement is evaluated later against available capacity. The separation is substantive: weak evidence, a binding quantity limit, and a later shortfall are different failures and should not be collapsed into one score.

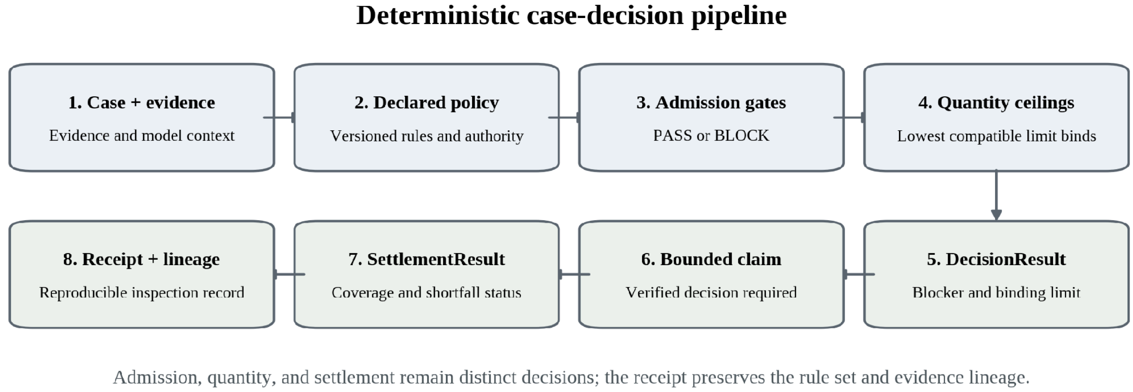

Figure 5.3. Deterministic case-decision pipeline in the V2 research workbench.

Table 5.3 maps the five constraints to these stages. Governance appears through a named and versioned policy with declared authority. The software does not eliminate discretion; it makes the chosen rule set, inputs, and consequences visible enough to challenge.

Table 5.3. Mapping the five constraints to deterministic decision stages.

Constraint	V2 decision representation	Question answered
Reliable energy data	Evidence, provenance, and context identity	Observed evidence or modelled context?
Rule-bound issuance	Admission gates and verified DecisionResult	May the claim proceed?
Explicit pricing and risk controls	Declared calculators and comparable quantity ceilings	What limits the permitted quantity?
Protected settlement	Separate settlement constraint and SettlementResult	What is covered and what remains short?
Limited governance	Versioned policy, manifest hash, declared authority	Which rule set produced the decision?

5.11 Controlled Case Workbench Evidence

The controlled case pack tests whether the architecture produces stable explanations rather than merely executable code. TYN-001, AUS-001, and PHX-001 use fixture evidence and model-based PVWatts/TMY context, not operator data.

Each committed ContextManifest preserves the provider, exact NSRDB weather-source string, coordinates, TMY semantics, annual AC output, 10 kW system size, and SHA-256 context hash. The manifests also state that modelled context cannot authorise a claim.

Taoyuan uses the Himawari tmy-2020 3.2.0 source; Austin and Phoenix use GOES tmy-2020 3.2.0 (National Laboratory of the Rockies, n.d.).

The case design tests four properties. It asks whether weak evidence blocks admission, whether different compatible ceilings can bind, whether a declared counterfactual changes the decision without changing the evidence file, and whether the result can be reproduced.

Taoyuan provides the clearest admission test. ENERGY-CASE-PILOT-005 requires at least L2 assurance, while the initial case declares L0. The result is BLOCKED by MIN_PROVENANCE and quantity is not calculated. In the deliberate L2 counterfactual, the evidence file and hash remain unchanged while only the declared assurance state changes. Admission then passes, the maximum quantity is 126 claim units, and the provenance-policy ceiling binds. The counterfactual changes the decision context, not the underlying evidence.

Austin and Phoenix show that the same engine can identify different binding constraints. AUS-001 is admitted at 283.09811 claim units because the resource-context ceiling is lowest. PHX-001 is admitted at 320 because the evidence-backed ceiling binds. The comparison does not rank locations; it shows that the decision is explainable in terms of the declared policy and inputs.

Table 5.4. Deterministic outcomes in the controlled V2 case pack.

Case / policy state	Decision	Maximum allowed (claim units)	Rule that blocked or limited the case
TYN-001 / L0 / pilot	BLOCKED	Not executed	MIN_PROVENANCE
TYN-001 / L2 / pilot	ADMIT WITH LIMIT	126	PROVENANCE_POLICY
AUS-001 / L2 / pilot	ADMIT WITH LIMIT	283.09811	RESOURCE_CONTEXT
PHX-001 / open	ADMIT WITH LIMIT	320	EVIDENCE_BACKED

Settlement produces a separate result. Declared capacity of 100%, 40%, and 0% yields SETTLED, PARTIAL, and SHORTFALL. For the 126-unit Taoyuan claim, the 40% replay records 50.4 units covered and 75.6 in shortfall. Admission and quantity discipline therefore reduce one class of risk without guaranteeing later performance.

5.12 Decision Receipts, Lineage, and Research Reproduction

A DecisionReceipt preserves the decision, inputs, governing policy, blocker or binding limit, source revision, and data boundary. Provenance is treated as part of the result: the system records not only what it decided but how that decision was produced (Moreau & Missier, 2013).

The interface organises inspection around Cases, Compare, Studies, Receipts, and Reference. Reviewers can trace constraints, evidence, stress, lineage, and the 3 × 3 case-policy comparison, while a research capsule packages the same record for later reproduction. These tools add no economic claim; they expose blockers, binding limits, counterfactual changes, and source boundaries.

At the frozen reviewed revision, the core, frontend, production build, browser walkthrough, CI, Solidity, security, and secrets checks passed. Appendix B reports the exact revision and test counts. The validation establishes internal consistency of the implementation, not operator validation or commercial readiness.

5.13 Implementation Conclusion

The two proof-of-concept layers answer different questions. SPK v1 shows that selected evidence, issuance, payment, redemption, and launch-gate rules can execute on a public testnet. The case workbench shows why a claim is admitted or blocked, what limits its quantity, how settlement diverges from admission, and whether the decision can be reproduced.

Under controlled assumptions, the five-constraint architecture can therefore be represented as an auditable decision and settlement process. The result is constructive but bounded: modelled context does not become proof of production, software does not create legal redemption rights, and successful tests do not establish deployment readiness.

Chapter 6 - Conclusion

6.1 Answer to the Research Question

Energy can constrain a digital financial claim, but only conditionally. Expenditure or production supplies a physical reference; it does not by itself determine market value, authorise issuance, create settlement rights, or limit governance. Credibility arises when evidence, quantity, risk, settlement, and authority are joined in rules that can be inspected and enforced.

The thesis therefore rejects both the passive and rhetorical versions of energy backing. Bitcoin's mining expenditure is not a redeemable reserve, the Chapter 4 benchmark is not an observed market price, and the testnet and controlled-case artifacts are not ready to hold public financial value. The result is an architecture for disciplined claims, not a proposal to replace fiat money.

6.2 Findings

The evidence follows a cumulative sequence. Chapter 3 reproduces the CEIR association and then shows why it does not identify an energy-specific anchor. Chapter 4 converts declared renewable-output uncertainty into inspectable financial quantities but shows that a number has no force until policy uses it. Chapter 5 then implements the missing rule chain: evidence determines admission, compatible constraints determine quantity, settlement capacity is tested later, and the decision remains traceable to a versioned policy.

6.3 Contributions

1. Conceptually, the thesis distinguishes physical reference from financial constraint. Energy occupies a possible middle position between commodity backing, fiat institutional credibility, and protocol scarcity, but only when its evidentiary and settlement consequences are explicit.
2. Empirically, the thesis converts a positive-looking result into a negative identification finding. The CEIR association is reproducible, yet the failed price merge, initial-stock sensitivity, negative controls, time-series diagnostics, and robust break test prevent an energy-price interpretation.
3. Methodologically, the thesis provides a transparent cold-start route from renewable-output uncertainty to option values, margin stress, and oracle-tolerance quantities, while preserving the distinction between scenario analysis and market evidence.
4. Architecturally, the thesis defines five jointly necessary constraints - evidence, issuance, pricing, settlement, and governance - and separates admission, quantity, and later performance as distinct financial decisions.

5. Technically, the thesis demonstrates that those decisions can be represented in software. Public-testnet contracts execute selected attestation, issuance, payment, and redemption rules; the deterministic workbench preserves evidence-policy lineage, binding constraints, settlement outcomes, and reproducible receipts.

6.4 Limitations

1. The Bitcoin evidence is limited to one proof-of-work asset and a panel whose market capitalisation is derived from a local price export and approximate supply curve. Mining electricity use is modelled rather than metered. The legacy CEIR denominator also contains a near-constant electricity-price path, a pre-sample 2018 cost stock, and highly persistent series. These defects are central to the negative result, but they prevent a clean test of a correctly constructed geography-weighted cost measure.
2. The 20 June 2021 split is researcher-defined and aligned with the broader mining crackdown rather than supplied as an official Cambridge event date. The robust joint test does not identify a reliable break, so geography, cost dispersion, and coordination remain hypotheses rather than established mechanisms.
3. The pricing chapter uses a cold-start scenario model. Real power markets may exhibit jumps, negative prices, mean reversion, seasonality, curtailment, basis risk, counterparty risk, and regulatory constraints that the GBM benchmark does not capture.
4. Public irradiance and production models support calibration and context, not proof of actual generation, export, or final settlement. Operator-grade measurement remains necessary for a live claim.
5. The implementation remains controlled proof-of-concept evidence. Sepolia cannot establish production cybersecurity, legal enforceability, demand, reserve sufficiency, tax treatment, utility compliance, or physical redemption. The V2 cases are fixtures, and the current policy pack does not convert the pricing benchmark into a validated risk-based quantity rule.

These limitations define the boundary of the contribution: the thesis demonstrates an inspectable research architecture, not a deployable financial product.

6.5 What Would Strengthen or Weaken the Thesis

The architecture is falsifiable. It would be weakened if real operator evidence cannot be classified consistently or if risk outputs cannot become defensible policy limits. Unstable binding explanations, ambiguous settlement rights, or governance that can waive the constraint when costly would weaken it further.

Those tests matter more than rescuing the retired CEIR anchor interpretation. A correctly rebuilt mining-cost panel could revise the empirical boundary, but the constructive architecture should stand or fail on whether it disciplines actual claims.

The relevant performance standard is therefore institutional: does the framework reduce discretion, expose assumptions, and make admission, quantity, and settlement easier to challenge and reproduce?

6.6 Future Research

1. Empirical work should rebuild the Bitcoin cost panel from a documented electricity-price series and extend the test to additional proof-of-work assets. Negative controls, seed treatment, and break specifications should be pre-declared.
2. Pricing work should replace the cold-start benchmark where data permit, incorporating mean reversion, jump diffusion, seasonality, curtailment, negative prices, and multi-factor power-market dynamics. The next contribution is not a more complicated model by itself, but a defensible rule that maps the output into compatible quantity or collateral units.
3. Implementation research should test one pre-declared pilot with real meter-level operator evidence before any claim carries public financial value. The case should preserve the operator, data source, measurement window, provenance state, policy version, risk rule, admission result, quantity limit, settlement outcome, and receipt. Governance boundaries, independent oracle checks, security review, audited contracts, and written settlement terms would remain prerequisites.

The pilot question should be deliberately narrow: for one declared evidence set and policy, what blocks the case or determines the maximum quantity?

If a claim is created, the second question is whether it settles as declared. Recording both questions before observing the outcome would allow the workbench to study which constraints bind and what predicts shortfall.

6.7 Closing Statement

The thesis ends with one decision rule.

An energy-linked claim should not exist unless its evidence, governing policy, quantity limit, settlement obligation, and decision lineage can be inspected and enforced.

Energy becomes credible not when a system says that energy was spent, but when the system cannot create or settle claims beyond the limits that its evidence and rules can support.

Appendix A - Supplementary Empirical Checks

A.1 Differenced CEIR boundary check

Table A.1. Differenced CEIR boundary check.

Item	Value
Pre-split β (Δlog CEIR)	-0.2357 (p=0.424)
Post-split β (Δlog CEIR)	0.1424 (p=0.378)

Differencing removes the significant CEIR effect. This is a boundary condition and supports the conclusion that the level result is driven by persistent series.

A.2 Trading-rule negative result

Table A.2. CEIR trading-rule comparison.

Metric	Value
Strategy total return (%)	176.4
Buy-and-hold total return (%)	2771
Sharpe (strategy)	0.723
Sharpe (buy-and-hold)	1.132

The CEIR rule underperforms buy-and-hold. CEIR is discussed as explanatory evidence, not as a usable trading strategy.

Appendix B - Implementation Evidence

B.1 Canonical Sepolia contracts and reviewed runtime snapshot

Source: frontend/public/spk_v1.json. The runtime file identifies Sepolia (chain ID 11155111) and records a sync from on-chain state at 2026-06-10T16:45:22Z. The figures below are a reviewed snapshot at that sync time, not live-current chain statistics.

Table B.1. Canonical Sepolia contract addresses.

Contract	Address
mock_usdc	0xaD2A7169CfFBA9Bef8C45515fc85178DbBfEc2C9
solar_punk_coin	0x8e189002228Fd4C6fA7611bA49FBe1d9C3412128
currency_system	0x520162252F9B94824417678525FFd69145014970

- Total supply: 5499.015 SPK
- Settled: 442.0 SPK
- Network payments: 21
- Circulation share: 96.71%

B.2 Selected on-chain activity

The reviewed Sepolia record contains 21 network payments. Payment #15 is the 5 SPK wallet-initiated pilot transfer. Payment #3 is a 180 SPK operator test action recorded in NETWORK form. These transactions are implementation evidence, not external commercial activity.

B.3 Deterministic V2 research artifact and validation

The case-workbench implementation was reviewed at exact source revision eb8714a6544b3480226283a69d41b3946df63451 on branch agent/case-workbench-v2. At that revision, the deterministic core passed 60/60 tests and the frontend passed 64/64 tests.

The SDK package-content check, production Vite build, and 15-state Chromium desktop/mobile walkthrough also passed.

The same review passed Case Workbench V2 CI, Constraint Protocol Alpha CI, Tests & Coverage, Solidity tests, Solidity security, and Secrets Scan.

PR #5 later merged after the stacked publication sequence was resolved. Revision eb8714a6544b3480226283a69d41b3946df63451 remains the frozen reviewed snapshot, and the reference outcomes remain tied to it.

The reviewed branch was a stacked research implementation. It did not deploy a new general protocol contract, change the existing SolarPunk/SPK deployment, edit the empirical files, or obtain real operator evidence. The validation shows that the implementation was internally consistent at the reviewed revision. It does not show production readiness or operator validation.

B.4 Controlled case pack reference outcomes

The controlled case pack contains TYN-001, AUS-001, and PHX-001. At the reviewed revision, TYN/L0/pilot is BLOCKED by MIN_PROVENANCE and quantity is not calculated. The TYN/L2 counterfactual is admitted at 126 ENERGY_CLAIM_UNIT, with PROVENANCE_POLICY_CAPACITY as the lowest limit.

AUS/L2/pilot is admitted at 283.09811, with RESOURCE_CONTEXT_CAPACITY binding. PHX under the open policy is admitted at 320, with EVIDENCE_BACKED_CAPACITY binding.

Settlement is tested in a separate stage. Declared settlement capacity of 100%, 40%, and 0% produces SETTLED, PARTIAL, and SHORTFALL respectively. For the 126-unit Taoyuan L2 case at 40% capacity, 50.4 units are covered and 75.6 remain in shortfall.

B.5 Decision identity, receipts, and reproduction

The decision ID is determined by the declared case, evidence, context, policy, and calculator inputs. Evaluation time is stored in the DecisionReceipt rather than in the decision hash. In the Taoyuan L0→L2 counterfactual, the evidence hash stays the same but the decision ID changes because the declared assurance state changes. Before a V2 claim is accepted, the software recomputes and verifies decision_id.

The DecisionReceipt records the case and decision IDs, policy version and policy hash, evidence hashes, context identities, evaluated rules, blockers, binding limits, final result, source revision, and data boundary. The static research-capsule export is designed to preserve the same inspection record for a case run.

Reproduction uses the core and frontend test commands, followed by the Vite build and the case-workbench capture script. The reviewed walkthrough covers 15 desktop and mobile states, including blocked and admitted cases, comparison, stress, lineage, receipts, and compatibility/reference routes.

B.6 Source-lineage audit

Table B.1 records what the audit could trace from external origin to local artifact and what remains unresolved. The purpose is not to claim perfect raw-data replay. It is to prevent a derived, modelled, or weakly documented input from being described as a directly observed series.

Table B.2. Source-lineage audit and unresolved provenance boundaries.

Input / evidence	**Audit result and boundary**
Bitcoin price and Market_Cap	Local BTC price export is preserved, but upstream vendor/request metadata are not. The builder uses Open as Price and an approximate supply curve, so Market_Cap is derived. The earlier CoinGecko label was removed.
CBECI electricity use	Cambridge best-guess annualised-TWh model export is preserved. This is modelled network electricity use, not direct miner metering.
Legacy electricity price	The intended country-year price source table is missing and the monthly-to-daily merge failed. The canonical path is near-constant; the energy-price-specific interpretation is retired.
Mining geography / event split	Monthly Cambridge mining-pool geolocation shares are preserved for Sep 2019–Jan 2022 and were manually structured locally. The 20 Jun 2021 split is researcher-defined, not a Cambridge event date.
Fear & Greed control	Alternative.me daily values are preserved locally. The raw request timestamp and checksum are not preserved.
NASA POWER transform	The Taiwan transform diagnostic is preserved. The exact NASA POWER parameter code and UTC/LST request setting are not source-locked in the audited artifacts.
S_0, r, and non-Taiwan σ	These are hard-coded scenario inputs. A complete dated external source ledger and non-Taiwan calibration artifacts are not preserved, so the thesis no longer describes them as observed market inputs.

Input / evidence	Audit result and boundary
V2 PVWatts contexts	Provider, exact NSRDB weather-source string, coordinates, TMY semantics, annual AC, 10 kW system size, SHA-256 hash, and the modelled-context boundary are preserved in committed manifests.
V2 case decisions	The frozen reviewed head eb8714a... and committed case, policy, and receipt artifacts preserve the reference outcomes. The cases remain controlled fixtures, not operator evidence.
SPK v1 runtime	The Sepolia runtime JSON preserves chain ID, contracts, deploy transactions, synced_at, and on-chain snapshot fields. The reported metrics are dated snapshot values, not live-current statistics.

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